

Naive Bayes Algorithm

$$P(+)=\frac{1}{2}$$

"spam filtering"

Bayes Theorem

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A

given B

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \implies P(A \cap B) = P(B|A) * P(A)$$

Likelihood

Prior

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

Posterior

Evidence

very often

Likelihood

Prior

$$P(\text{spam} | \text{"winning"}) \implies \frac{P(\text{"winning"} | \text{spam}) * P(\text{spam})}{P(\text{"winning"})}$$

Final
Outcome

$$\approx 0.49$$

P("winning")
Evidence

$$P(\text{ham} | \text{"winning"}) = \frac{P(\text{"winning"} | \text{ham}) * P(\text{ham})}{P(\text{"winning"})}$$

$$\approx 0.52$$

P("winning")

Independence Assumption



→ every feature is independent

Naive

of every other

↳ just to simplify

feature

the maths

↳ realistically the

assumption will not

hold true

Types of Naive Bayes

① Gaussian Naive Bayes (GaussianNB)

↳ height, weight, temp

Input

features

α (normal distribution)

(decimal)

Medical data

② Multinomial Naive Bayes } Discrete

↳ count (integer)

text classification → count how many num

of times words appear

in a given

document

Message $\rightarrow$ spam or ham

"Win a free lottery tickets now"

spam

CountVectorizer Vectors (numeric mapping of the given words)

Model Training

text = {
hello, how are you
I hope, you are okay
something okay

unique words

	hello	,	how	are	you	I	hope	okay	something
0	1	1	1	1	1	0	0	0	0
1	0	1	0	1	1	1	1	1	0
2	0	0	0	0	0	0	0	1	1

$\hookrightarrow$ count of occurrence of a given word

	<u>outlook</u>			
	yes	no	P(yes)	P(no)
sunny	2	3	2/9	3/5
overcast	4	0	4/9	0
rainy	3	2	3/9	2/5
	9	5		

f_1, f_2, f_3, f_4

	outlook	temperature	humidity	wind	play
1	sunny	hot	high	FALSE	NO
2	sunny	hot	high	TRUE	NO
3	overcast	hot	high	FALSE	YES
4	rainy	mild	high	FALSE	YES
5	rainy	cool	normal	FALSE	YES
6	rainy	cool	normal	TRUE	NO
7	overcast	cool	normal	TRUE	YES
8	sunny	mild	high	FALSE	NO
9	sunny	cool	normal	FALSE	YES
10	rainy	mild	normal	FALSE	YES
11	sunny	mild	normal	TRUE	YES
12	overcast	mild	high	TRUE	YES
13	overcast	hot	normal	FALSE	YES
14	rainy	mild	high	TRUE	NO

→ classification task

→ Yes/No

$$P(\text{Yes} \mid \text{sunny, hot, normal, true})$$

$$P(\text{No} \mid \text{sunny, hot, normal, true})$$

$$= \frac{P(B|A) * P(A)}{P(B)}$$

← sunny) ← normal true
 ↓ hot
 new test data

$$\Rightarrow P(\text{sunny} | \text{yes}) * P(\text{hot} | \text{yes}) * P(\text{normal} | \text{yes}) * P(\text{true} | \text{yes}) * P(\text{yes}) \rightarrow 9/14$$

(D) ~~$P(\text{sunny}) * P(\text{hot}) * P(\text{normal}) * P(\text{true})$~~

ignore this denominator

$$\Rightarrow 0.007$$

	temp		P(yes)	P(no)
	yes	no		
hot	2	2	2/9	2/5
mild	4	2	4/9	2/5
cool	3	1	3/9	1/5
	9	5		

	humidity		$P(\text{yes})$	$P(\text{no})$
	yes	no		
high	3	4	$3/9$	$4/5$
normal	6	1	$6/9$	$1/5$
	<u>9</u>	<u>5</u>		

	Wind		$P(\text{yes})$	$P(\text{no})$
	yes	no		
true	3	3	$3/9$	$3/5$
false	6	2	$6/9$	$2/5$
	<u>9</u>	<u>5</u>		

$$\Rightarrow P(\text{sunny} | \text{no}) * P(\text{hot} | \text{no}) * P(\text{normal} | \text{no}) * P(\text{false} | \text{no}) * P(\text{no}) \rightarrow 5/14$$

~~$$P(\text{sunny}) * P(\text{hot}) * P(\text{normal}) * P(\text{false})$$~~

$$\Rightarrow \underline{0.006}$$

$$P(\text{yes} | \text{feature}) > P(\text{no} | \text{feature})$$

$$\underline{\text{prediction}} \Rightarrow \underline{\text{yes}}$$